

QUANTITATIVE ECONOMICS - SPRING 2026

Answer any FIVE questions.

1. Consider the regression model

$$Y_i = \beta X_i + u_i \quad i = 1, \dots, n$$

where u_i and X_i satisfy the standard assumptions. Let $\bar{\beta} = \frac{\bar{Y}}{\bar{X}}$, where $\bar{Y}$ and $\bar{X}$ are the sample means of Y_i and X_i , respectively.

- (a) Show that $\bar{\beta}$ is a linear function of $Y_1, Y_2, \dots, Y_n$.
- (b) Show that $\bar{\beta}$ is conditionally unbiased.
- (c) Derive the variance of $\bar{\beta}$.

2. Consider the model

$$y = X\beta + e,$$

where β is a $k \times 1$ vector, $E(e | X) = 0$, and $V(e | X) = \sigma^2$. Suppose we are interested in the object $\theta = r(\beta) = R\beta$, where R is a $m \times k$ matrix and m denotes the number of linear restrictions.

- (a) Suppose we wish to test the null $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$. Write the matrix R .
- (b) Suppose we wish to test the null $H_0 : \beta_1 + \beta_2 + \dots + \beta_k = 0$. Write the matrix R .
- (c) Suppose we wish to test the null $H_0 : \beta_1 + 2\beta_2 = 0$. Write the matrix R .
- (d) Consider any one of the tests above. Write down the Wald statistic for this null.
- (e) What is the distribution of your test statistic from part (d)?

3. Consider the model

$$y = X\beta + e,$$

where $E(e | X) = 0$ and $E(ee' | X) = \Omega$, where Ω is an unknown variance-covariance matrix.

- (a) Derive the OLS estimator $\hat{\beta}$ of β .
- (b) Show that the estimator from part (a) is unbiased.
- (c) Using the result from part (a), propose an estimator $\hat{\Omega}$ for the variance-covariance matrix Ω .
- (d) Using the result from part (c), and the objective function $\tilde{e}'\hat{\Omega}^{-1}\tilde{e}$, derive the feasible generalized least squares estimator $\tilde{\beta}$ of β .
- (e) Show that the estimator from part (d) is unbiased.

4. Consider the R output listed below. With this information, answer the following:

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model.fit = arima(data,order=c(0,0,2),method='ML')
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model.fit
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Coefficients:
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	ma1	ma2	intercept
	0.4705	0.2496	-0.0190
s.e.	0.7376	0.3428	1.7343

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sigma2 estimated as 40.51: log likelihood = -140.76, AIC = 291.53, AICc = 293.15, BIC = 300.33
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- Find the h -step-ahead forecast for $h = 1, 2, 3$.
 - Find the h -step-ahead forecast error for $h = 1, 2, 3$.
 - Find the h -step-ahead forecast error variance for $h = 1, 2, 3$.
 - Find the h -step-ahead forecast interval for $h = 1, 2, 3$.
 - Plot parts (a) and (d) in a single figure.
5. In 2010, City A and City B were two neighboring cities with similar population demographics and public health metrics. In 2011, City A launched a public health campaign focused on reducing smoking rates through awareness programs and free cessation aids, while City B continued its existing policies without implementing additional interventions. Researchers collected data on the smoking status of residents from a random sample in both cities in the years before and after the campaign's implementation to investigate its effects on the likelihood of individuals quitting
- Using a difference-in-differences (DiD) identification strategy, formulate a linear probability model (LPM) to examine whether the public health campaign in City A in 2011 affected the probability of individuals quitting smoking by 2012. Identify the coefficient in your model that quantifies the effect of the public health campaign.
 - Interpretation of the Coefficient of Interest: How would you interpret the coefficient associated with the public health campaign in your LPM?
 - Control Variables: Suggest potential control variables to include in the model. Explain how these variables might help address omitted variable bias and improve the precision of your estimates.
 - Parallel Trend Assumption: Define the Parallel Trend Assumption in the context of this study. What strategies or methods could you use to assess whether this assumption is met, utilizing data from residents in City A and City B?
6. In 2015, a state introduced a program to encourage energy savings. Households with an income below \$50,000 could qualify for a tax reimbursement if they installed energy-saving appliances. However, some households slightly above \$50,000 also received the reimbursement, and some below \$50,000 did not. Researchers want to study the impact of receiving reimbursement on household energy use in the following year using a regression discontinuity (RD) design.
- Sharp vs. Fuzzy RD Design: Explain the difference between a sharp and a fuzzy regression discontinuity design. Clearly write the discontinuity condition for both types of RD designs.
 - Appropriate RD Design: Which type of RD design (sharp or fuzzy) is more appropriate for this study? Justify your answer based on how the reimbursement was distributed and write down the regression formula that corresponds to the chosen RD design.
 - 2SLS Estimation Strategy: Does the RD model in this context require a two-stage least squares (2SLS) approach? If yes, write the key equations and explain the first stage, reduced form, instrument, and instrumented variables. If no, explain why not.

7. A researcher is studying the impact of internet access (S_i) on household income (Y_i) across different regions. To address potential endogeneity issues, the researcher uses the distance to the nearest internet service provider hub (Z_i) as an instrumental variable. The researcher also includes control variables (X_i), such as education levels and regional employment rates, to account for other factors that might influence household income.
- (a) Write the first stage and reduced form equations for the instrumental variables (IV) model, where the aim is to estimate the causal impact of internet access (S_i) on household income (Y_i) using the distance to the nearest internet service provider hub (Z_i) as the instrument, along with covariates (X_i).
 - (b) Demonstrate that the IV estimator in your model can be expressed as a function of two coefficients obtained from the first stage and reduced form equations.
 - (c) Derive the reduced form equation for a Two-Stage Least Squares (2SLS) model by utilizing the first stage equation and the causal relation of interest mentioned above.
 - (d) What are the two key requirements for a good instrumental variable? Do you think the distance to the nearest internet service provider hub satisfies these requirements? Explain your reasoning.

8. Suppose we have $t = 1, 2, \dots, T$ observations on y_t and believe the model is of the form

$$y_t = c + \delta t + \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

where $0 < \phi < 1$ and ε_t is a white noise sequence.

- (a) What is the expected value of this series?
- (b) Use differencing to make this series stationary.
- (c) Instead of differencing, show how you can detrend the series.
- (d) For the detrended series obtained in part (c), what is the expected value of this series?
- (e) For the detrended series obtained in part (c), forecast the series for $h = 1, 2, 3$.